

1. Open your own laptop or desktop (or use an online PC hardware simulator if you don't have access to a real system) and identify the GPU model installed. Write down the GPU's brand, model number, and whether it is integrated or dedicated.

ANS:- My laptop has an **NVIDIA GeForce RTX 3050 6GB Laptop GPU** installed. It is manufactured by **NVIDIA** and is a **dedicated (discrete) graphics card**, which means it has its own dedicated video memory (6 GB VRAM). This GPU is designed for gaming, video editing, 3D rendering, and other graphics-intensive tasks, providing much better performance than an integrated GPU.

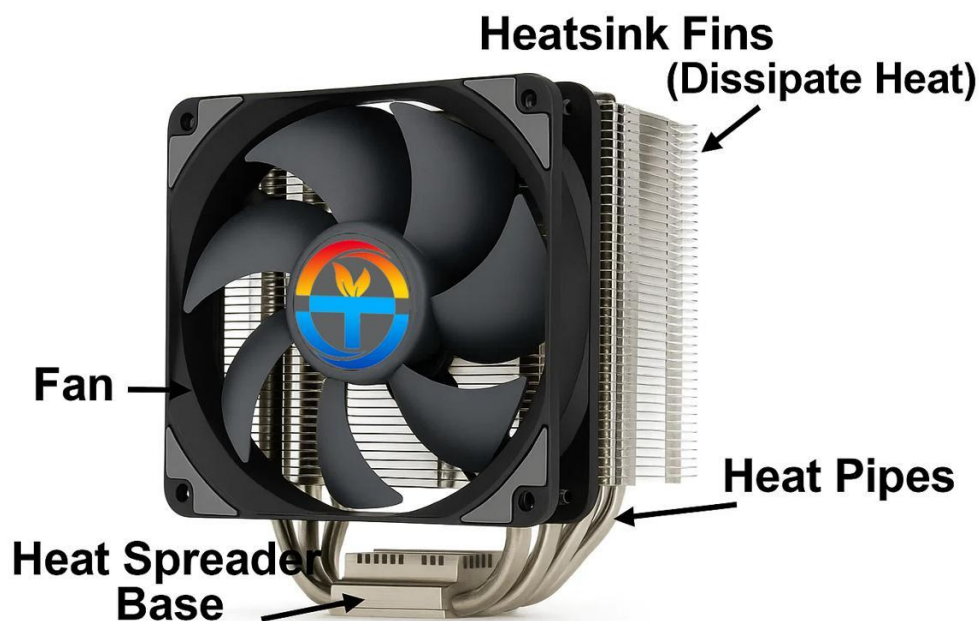
2. Take a clear photo or screenshot of the cooling system (fan, heatsink, or liquid cooling setup) inside your PC or from a PC hardware simulator. Label the main cooling components and briefly describe their function.

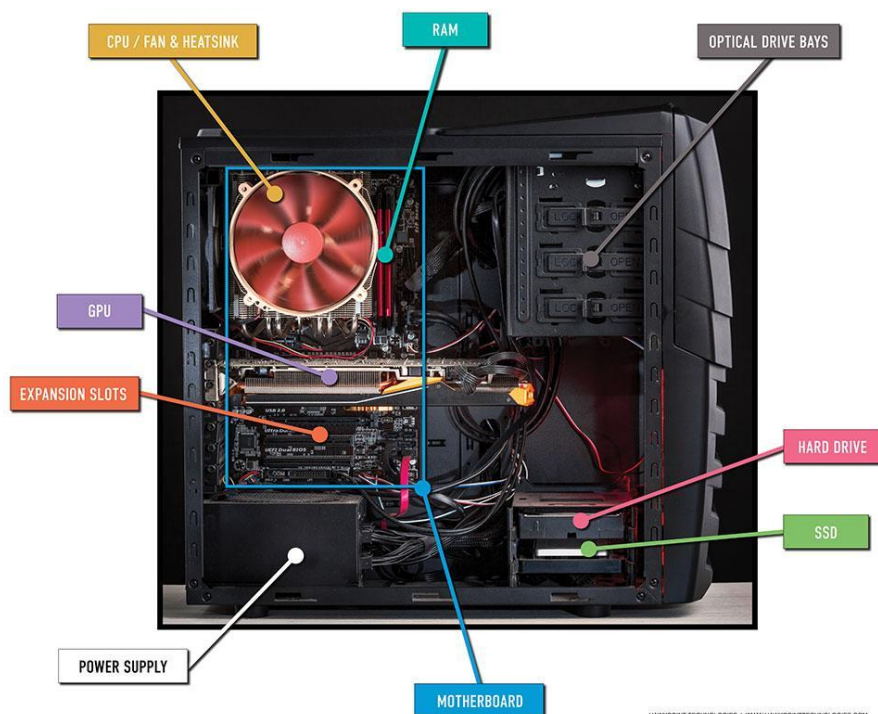
ANS:- Cooling System (Photo/Screenshot)

Use either:

- Your own PC (open the side panel safely after turning off the power), or
- An online PC hardware simulator.

Below are the components you should label:





Label these components

Component	Function
Cooling Fan	Pushes cool air over the heatsink to remove heat.
Heatsink	Absorbs heat from the CPU or GPU and dissipates it into the air.
Radiator (Liquid Cooling)	Releases heat from the coolant into the surrounding air.
Pump (Liquid Cooling)	Circulates coolant through the cooling loop.
Tubes (Liquid Cooling)	Carry coolant between the pump and radiator.

3. Create a comparison table listing at least 3 differences between air cooling and liquid cooling systems for GPUs, including factors like cost, efficiency, and maintenance.

ANS:-

Feature	Air Cooling	Liquid Cooling
Cost	Lower	Higher
Cooling Efficiency	Good for most users	Excellent for high-performance systems
Maintenance	Very low	Requires more maintenance and monitoring
Installation	Easy	More complex
Noise	Can become louder under heavy load	Usually quieter during heavy workloads
Best For	Budget and mid-range gaming PCs	High-end gaming, streaming, and overclocking

4. Imagine you are building a gaming PC for streaming IPL matches and playing graphics-heavy games. Choose between air cooling and liquid cooling for your GPU, and justify your choice in 3-4 sentences based on your research.

ANS:-

If I were building a gaming PC for streaming IPL matches and playing graphics-heavy games, I would choose **liquid cooling** for the GPU. It keeps the GPU cooler during long hours of gaming and streaming, which helps the system perform smoothly without overheating. It also runs more quietly than most air coolers when the PC is under heavy load. Even though liquid cooling is more expensive and takes a little more effort to install, I think it is worth it for better performance and reliability.